

Hayat Unified Hospital App



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Abstract

Healthcare delivery in Saudi Arabia is characterized by a fragmented landscape where services are distributed across multiple independent hospital systems. This lack of integration forces patients to navigate disparate applications to access appointments, medical records, and diagnostic data, ultimately hindering operational efficiency and degrading the patient experience. This project presents Hayat, a centralized, patient-centered mobile solution designed to consolidate healthcare services from diverse providers into a unified digital ecosystem.

Developed using an Agile-iterative methodology, Hayat leverages modular system architecture to ensure high scalability, security, and seamless data integration. Beyond standard functionalities such as appointment scheduling and real-time access to laboratory results, prescriptions, and vaccination histories the application incorporates an AI-driven symptom analysis engine. This feature facilitates intelligent triage, guiding users toward the most appropriate medical interventions based on preliminary data.

The system's requirements were derived from a comprehensive gap analysis of current e-health platforms in the Kingdom. The implementation of Hayat demonstrates a robust application of Information Systems (IS) to bridge the gap in healthcare interoperability. By streamlining information flow and enhancing patient decision-making, Hayat supports the strategic objectives of the Saudi healthcare digital transformation. The final outcome provides a scalable framework for improving accessibility, reducing administrative fragmentation, and fostering a more cohesive national health informatics infrastructure.

الملخص

يشهد قطاع الرعاية الصحية في المملكة العربية السعودية تحولاً رقمياً متسارعاً، إلا أن العديد من أنظمة المستشفيات ما تزال تعمل بشكل مستقل، مما يؤدي إلى تشتت الخدمات الصحية وصعوبة وصول المرضى إليها عبر منصات متعددة. تتسبب هذه التجزئة في زيادة الوقت والجهد المطلوبين لحجز المواعيد أو الوصول إلى السجلات الطبية، خاصة في الحالات الطارئة. يهدف هذا المشروع إلى تصميم وتطوير تطبيق موحد يحمل اسم "حياة"، يدمج خدمات عدة مستشفيات ضمن منصة واحدة تتمحور حول احتياجات المريض.

يوفر التطبيق إمكانية البحث عن المستشفيات، حجز المواعيد، الوصول إلى الوصفات الطبية وسجلات التطعيم، إضافة إلى ميزة تحليل الأعراض باستخدام الذكاء الاصطناعي لتوجيه المستخدم إلى الخدمة الصحية المناسبة. تم تطوير النظام باستخدام منهجية التطوير التكراري (Agile)، مع التركيز على سهولة الاستخدام، تكامل الأنظمة، وأمن البيانات.

يتوقع أن يسهم تطبيق "حياة" في تحسين كفاءة الوصول إلى الخدمات الصحية، وتقليل تشتت الأنظمة، وتعزيز تجربة المريض. كما يبرز المشروع دور نظم المعلومات في دعم التحول الرقمي للقطاع الصحي من خلال تحقيق التكامل بين الأنظمة وتحسين تدفق المعلومات واتخاذ القرار.

Table of Contents

Hayat Unified Hospital App	1
Cover Page	2
Acknowledgments	3
Abstract	4
المخلص	5
Table of Contents	6
List of Figures	7
List of Tables	7
List of Acronyms	7
Chapter 1 – Introduction	8
1.1 Background	8
1.2 Problem Statement	8
1.3 Project Objectives	8
1.4 Scope and Constraints	9
1.5 Stakeholders and User Personas	9
1.5.1 Stakeholder Personas	10
1.6 Industry Relevance & IS Application	11
1.7 Project Timeline (Gantt Chart)	12
Chapter 2 – Literature Review	13
2.1 Overview of Previous Studies	13
2.2 Comparative Analysis	14
Chapter 3 – Methodology	16
3.1 Research Design / Project Approach	16
3.2 Data Collection / Requirements Gathering	16
3.3 System Design & Architecture	16
3.4 Implementation Strategy	20
3.5 Testing & Validation	20
3.6 Ethical, Legal & Security Considerations	20
Chapter 4 – Implementation and Evaluation	21
4.1 Implementation	21
4.2 Evaluation	25

4.3 Individual and Group Contributions	25
Chapter 5 – Conclusion and Future Work	28
References	29

List of Figures

Figure 1: Project Timeline Gantt Chart
11

Figure 2: Use Case Diagram
15

Figure 3: Activity Flow Diagram
16

Figure 4: Sequence Diagram
17

Figure 5: Hayat Workflow
19

Figure 6: Hayat Application image 1
20

Figure 7: Hayat using Morafiq
21

Figure 8: Hayat using Morafiq 2
22

Figure 9: Hayat using Morafiq 3
22

Figure 10: Hayat using Morafiq 4
22

List of Tables

Table 1: Project Tasks and Task Assignments
23

List of Acronyms

Chapter 1 – Introduction

1.1 Background

Digital transformation in healthcare has increased the availability of mobile health applications. However, many hospital systems operate independently, resulting in fragmented patient experiences. A unified information system is needed to streamline access to healthcare services.

1.2 Problem Statement

Despite the availability of national healthcare platforms such as Sehhaty, patients in Saudi Arabia still face challenges in accessing real-time appointment availability across multiple hospitals, particularly within the private sector. Existing systems primarily operate within isolated networks or focus on public healthcare services, lacking a centralized mechanism for cross-hospital booking and immediate availability matching. This limitation results in delays in accessing appropriate care, especially in urgent but non-emergency situations. Therefore, there is a need for a unified system that integrates multiple healthcare providers while enabling intelligent, real-time matching between patient needs and available healthcare services

1.3 Project Objectives

1- Design and develop a unified mobile application prototype that integrates at least three core healthcare services (hospital search, appointment booking, and medical record access) within a single platform by the end of the project.

2- Implement and evaluate an AI-based assistant (Morafiq) that analyzes user-input symptoms and provides relevant healthcare service recommendations, with its performance evaluated based on response efficiency and recommendation accuracy through system testing.

3-Ensure secure data handling by implementing authentication and data protection mechanisms, and validating the system through usability and security testing.

1.4 Scope and Constraints

The project focuses on developing a mobile application prototype integrating hospital search, appointment booking, and AI recommendations (Morafiq). Constraints include limited access to real hospital APIs, time limitations, and strict data privacy requirements.

1.5 Stakeholders and User Personas

The Hayat system involves several stakeholders who are directly or indirectly affected by the application. The primary stakeholder is the patient, who directly interacts with the system to access healthcare services. Patients use the application to search for hospitals, book appointments, and receive AI-based symptom guidance from Morafiq. The system is designed with a patient-centered approach to enhance convenience, accessibility, and efficiency.

Secondary stakeholders include:

- **Hospital Management and Administrators**, who provide appointment availability and healthcare service data to the system.
- **Healthcare Service Providers**, whose services are listed within the unified platform.
- **Healthcare IT and Security Teams**, who are responsible for maintaining system integration, data security, reliability, and compliance with healthcare data protection regulations.
- **Regulatory and Healthcare Authorities**, who benefit from improved interoperability and digital healthcare service efficiency.

By identifying these stakeholders, the system ensures alignment between patient needs, institutional requirements, and technical infrastructure.

1.5.1 Stakeholder Personas

Persona: Patient (Primary User)

Role: Healthcare Service User

Technical Skill Level: Basic to Moderate

Background:

The patient represents an individual seeking efficient access to healthcare services without navigating multiple hospital systems. He may require routine appointments, urgent consultations, or access to medical records.

Goals:

- Access healthcare services through a single unified platform
- Book appointments using Morafiq quickly and efficiently
- Identify appropriate medical services during urgent situations
- Access prescriptions and vaccination records in one place

Challenges:

- Fragmented hospital applications
- Delays caused by multiple booking systems
- Difficulty identifying appropriate services during emergencies

How Hayat Supports Him:

Hayat integrates multiple hospitals into a centralized platform, reducing system fragmentation. The AI-based symptom analysis called Morafiq assists him in identifying suitable medical services, while centralized access to appointments and records improves efficiency, accessibility, and overall healthcare experience.

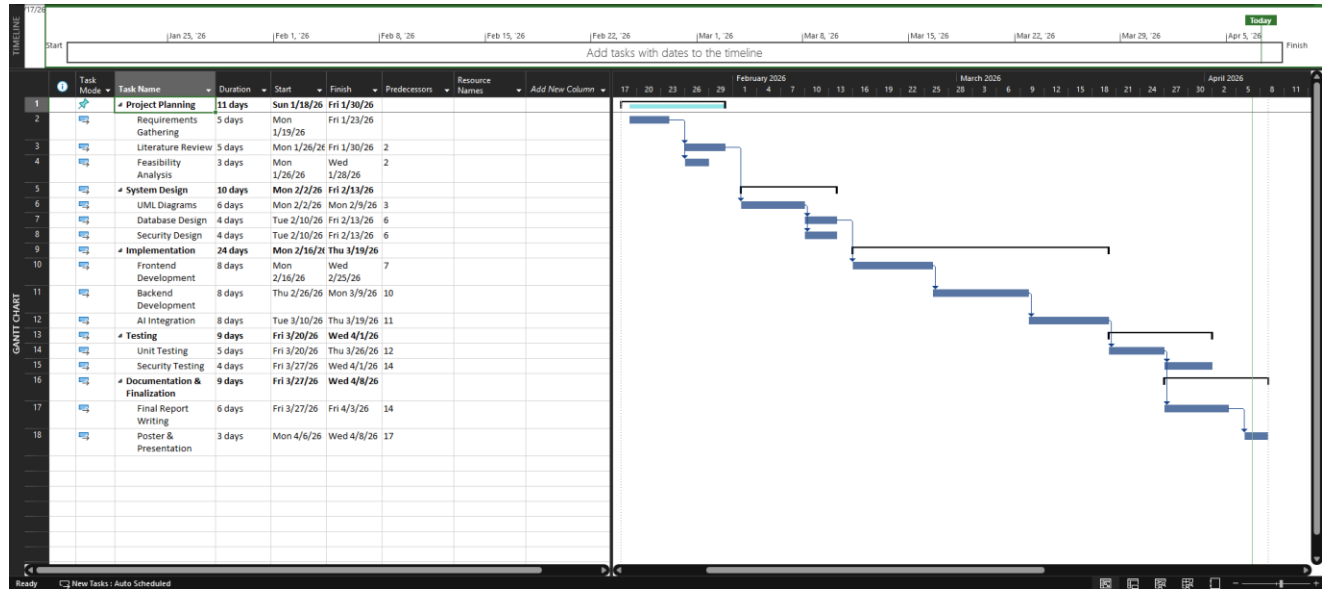
1.6 Industry Relevance & IS Application

The Hayat project addresses a major healthcare industry issue in Saudi Arabia: fragmented hospital information systems that force patients to use multiple applications to access essential services such as appointment booking, lab results, prescriptions, and vaccination records. This fragmentation increases effort, delays access to care, and reduces overall patient satisfaction, especially in urgent situations. Hayat is industry-relevant because it proposes a unified, patient-centered platform that simplifies healthcare access by integrating services from different hospitals into a single mobile application.

From an Information Systems perspective, Hayat applies key IS concepts including system integration, interoperability, centralized information management, and digital service delivery. Morafiq is The AI-based symptom guidance feature that supports patient decision-making by helping users identify the most suitable medical service and proceed with booking efficiently. Overall, the project demonstrates how Information Systems can improve healthcare service quality by enhancing information flow, reducing system fragmentation, and delivering a more efficient and accessible patient experience.

1.7 Project Timeline (Gantt Chart)

The following schedule presents a high-level breakdown of the project phases, their durations. This is further illustrated in the Gantt chart below.



Chapter 2 – Literature Review

2.1 Overview of Previous Studies

The adoption of Information Systems (IS) in healthcare has significantly transformed service delivery, operational efficiency, and patient engagement. Integrated Health Information Systems (HIS) have been widely studied for their ability to centralize patient data, improve coordination between healthcare entities, and enhance clinical and administrative decision-making processes. Research indicates that fragmented healthcare systems lead to inefficiencies, increased administrative workload, duplication of records, and reduced patient satisfaction, particularly in rapidly evolving healthcare environments such as Saudi Arabia under Vision 2030 [1]. As a result, interoperability and centralized data management have become major priorities in healthcare digital transformation initiatives.

Mobile health (mHealth) applications represent an important evolution of healthcare information systems. These applications enable patients to book appointments, access electronic medical records, receive prescriptions, and obtain healthcare guidance remotely. Studies demonstrate that mHealth solutions improve service accessibility, reduce waiting times, and enhance user convenience while supporting patient engagement in healthcare processes [2]. However, many mHealth systems operate within isolated institutional boundaries, limiting cross-hospital data exchange and integration.

Recent advancements in Artificial Intelligence (AI) have further enhanced healthcare information systems by introducing intelligent decision-support mechanisms. AI-based tools are increasingly used for symptom analysis, triage support, predictive modeling, and personalized healthcare recommendations [3]. These technologies contribute to improved resource allocation and more efficient patient routing. Despite these advancements, the integration of AI capabilities within unified, cross-institution healthcare platforms remains limited.

Recent research has also explored the use of Natural Language Processing (NLP) and Large Language Models (LLMs) in healthcare applications, particularly for symptom analysis

and patient triage. AI-driven conversational agents can interpret user input, assess symptoms, and provide preliminary healthcare guidance [4]. Studies further highlight that AI-based symptom checkers can improve early-stage triage accuracy while increasing accessibility to healthcare services, although concerns remain regarding reliability and clinical safety [5]. This highlights the importance of integrating AI-based symptom analysis with real healthcare services, ensuring that recommendations are actionable and connected to actual medical resources.

2.2 Comparative Analysis

Several digital healthcare applications operate in Saudi Arabia; however, each addresses only a specific component of healthcare service delivery rather than offering a fully integrated solution.

Sehaty primarily supports government healthcare services by enabling appointment booking within Ministry of Health facilities. While it improves access to public healthcare, it does not integrate private hospitals into a unified platform, limiting its scope of interoperability.

Tawakkalna functions mainly as a digital identity and public health verification system. Although it contributes to healthcare-related services, it does not operate as a comprehensive hospital integration system nor provide centralized healthcare service management.

Vezeeta allows patients to search for doctors and book appointments across different hospitals. Despite offering cross-institution appointment booking, it mainly acts as an appointment aggregation platform. It does not provide centralized access to medical records nor incorporate AI-driven symptom analysis within an integrated healthcare information system.

The common limitation among these systems is functional specialization without full system integration. Existing platforms focus on isolated services such as appointment

scheduling, identity verification, or specific institutional access. None provide a unified, patient-centered architecture that integrates public and private hospitals while simultaneously offering intelligent decision-support capabilities.

Globally, AI-based symptom checkers such as Ada Health and Babylon Health provide advanced diagnostic support using artificial intelligence and machine learning techniques. These platforms are effective in analyzing symptoms and suggesting possible medical conditions. However, they primarily function as standalone diagnostic tools and do not provide direct integration with local healthcare systems for appointment booking. In contrast, the Hayat system integrates AI-driven symptom analysis (Morafiq) with a unified healthcare platform, enabling users to not only receive recommendations but also immediately access relevant healthcare services and book appointments within the same system.

The Hayat system addresses this gap by providing a centralized platform that integrates healthcare services with intelligent decision support, improving accessibility, interoperability, and patient experience.

Chapter 3 – Methodology

3.1 Research Design / Project Approach

This project follows an implementation-based Information Systems approach using an Agile iterative development methodology. The project focuses on designing and developing a functional prototype of a patient-centered healthcare application rather than conducting experimental or statistical research. Agile methodology was selected due to its flexibility, support for incremental development, and ability to adapt to changing requirements. The iterative nature of Agile allows continuous refinement of system features based on feedback and evaluation at each development stage.

3.2 Data Collection / Requirements Gathering

System requirements were identified through the analysis of existing healthcare applications, review of similar systems, and examination of user needs described in Form A and Form B. Functional requirements include hospital search, appointment booking, access to medical records, and Morafiq's AI-based symptom guidance. Non-functional requirements focus on usability, performance, security, and data privacy. Requirements were documented and refined iteratively to ensure alignment with project objectives and patient needs.

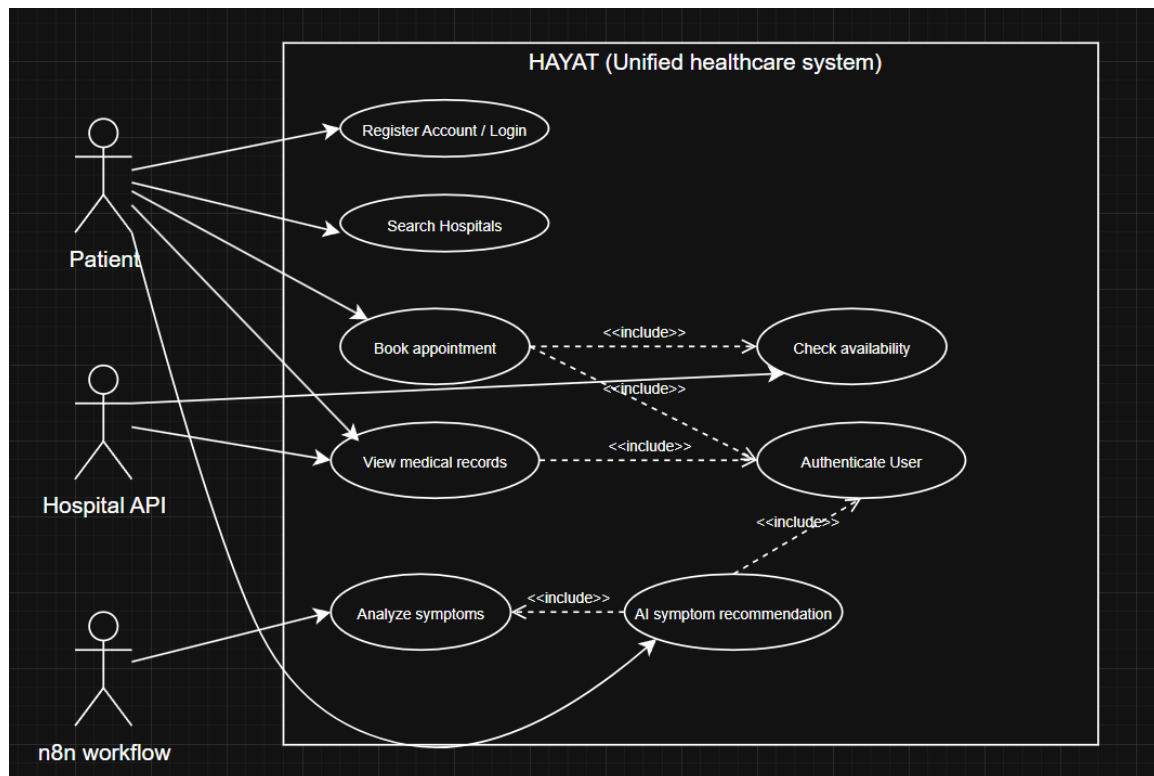
3.3 System Design & Architecture

The Hayat system follows a three-tier architecture consisting of the presentation layer, application layer, and data layer. The presentation layer (frontend) is developed using Flutter, providing a cross-platform mobile interface focused on usability and accessibility. The application layer (backend) is implemented using FastAPI, which handles business logic, API communication, authentication, and system integration. The data layer is

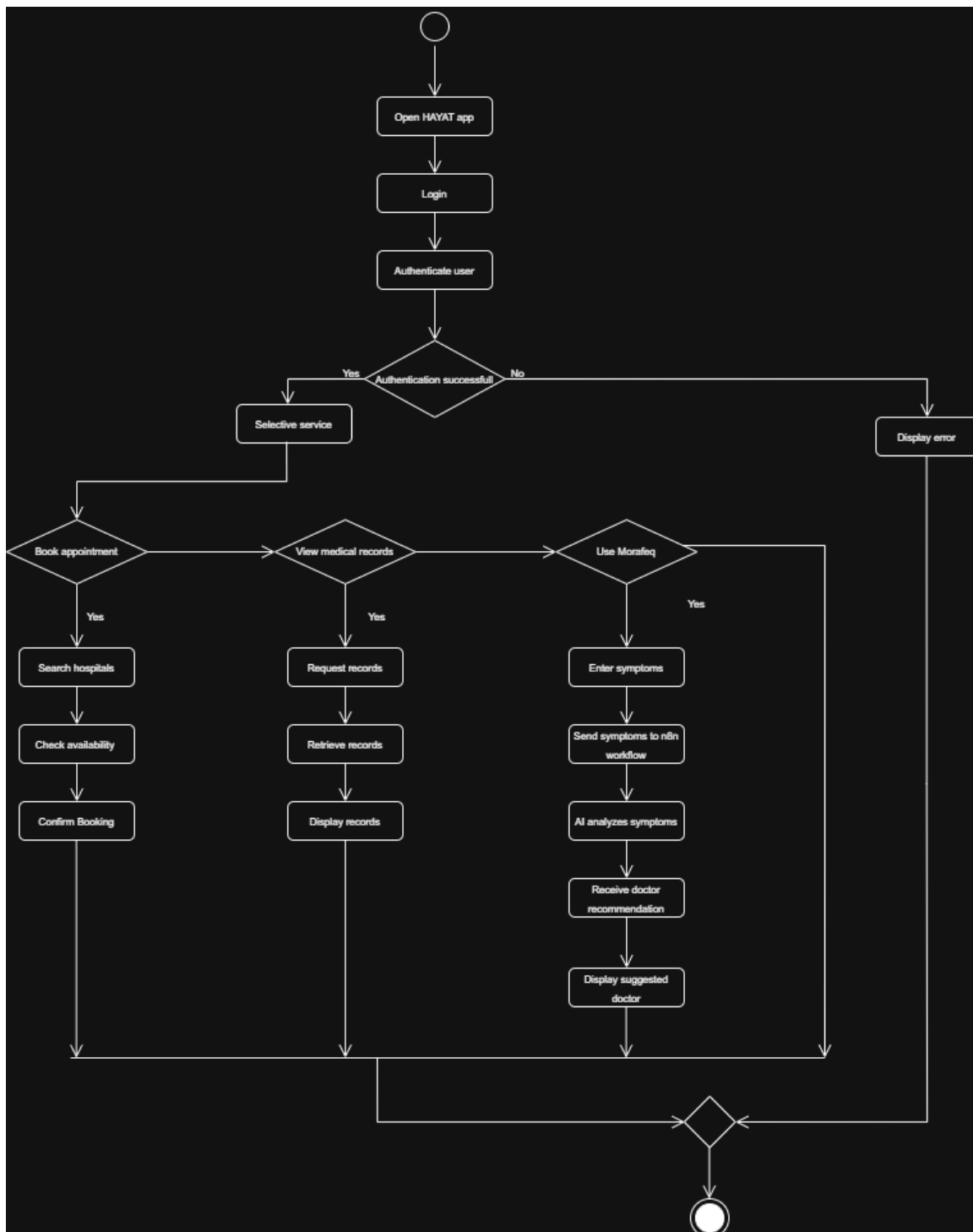
responsible for storing patient-related data, appointment information, and system records in a structured database.

In addition, the system integrates an AI component called Morafiq, which is based on a Large Language Model (LLM). This component utilizes Natural Language Processing (NLP) techniques to analyze user-input symptoms and generate appropriate healthcare recommendations. The architecture is designed to support scalability, modularity, and secure data exchange between system components.

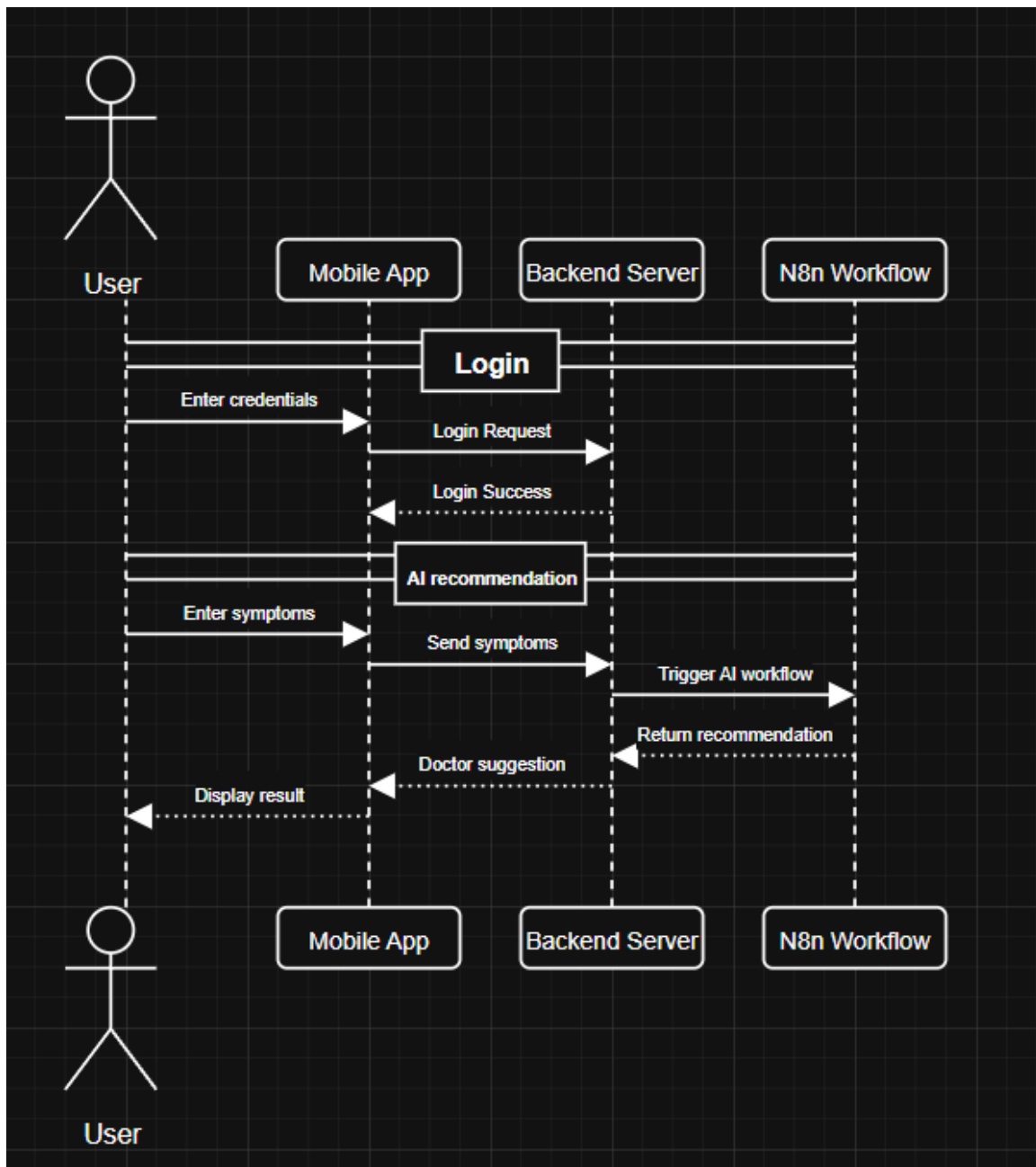
Use Case Diagram:



Activity Flow Diagram:



Sequence Diagram:



3.4 Implementation Strategy

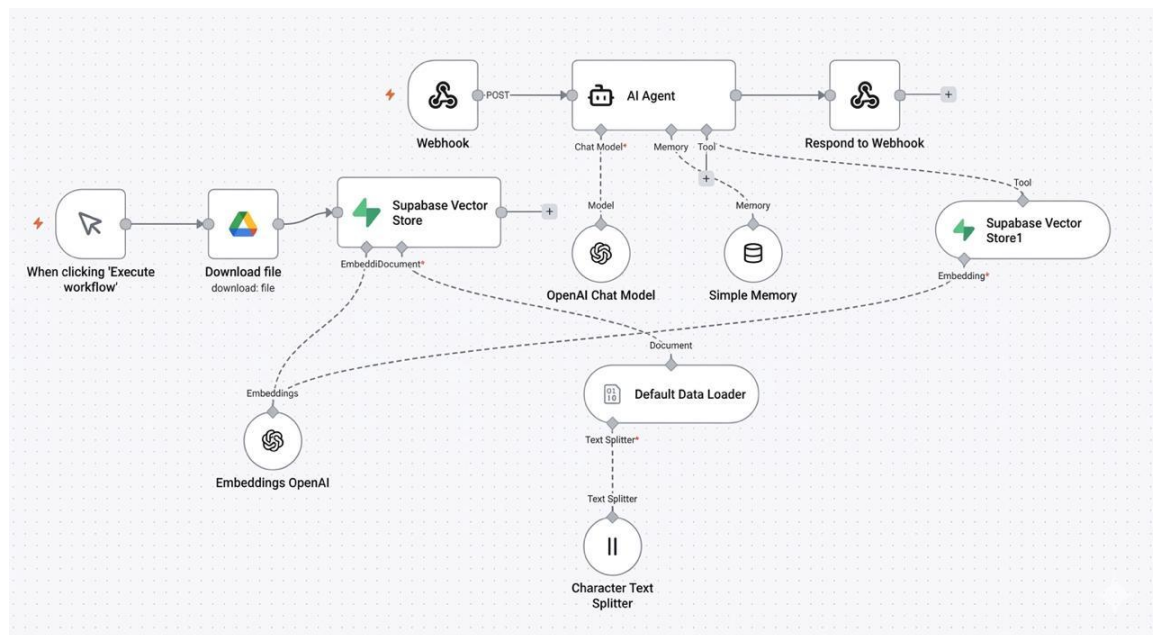
The implementation was carried out incrementally according to Agile principles. Development tasks were divided into iterations, with each iteration focusing on a specific system feature. Version control tools were used to manage source code and track changes throughout development. The implementation strategy emphasized usability, secure data handling, and reliable system performance. Regular reviews were conducted to ensure that implemented features met the defined requirements.

3.5 Testing & Validation

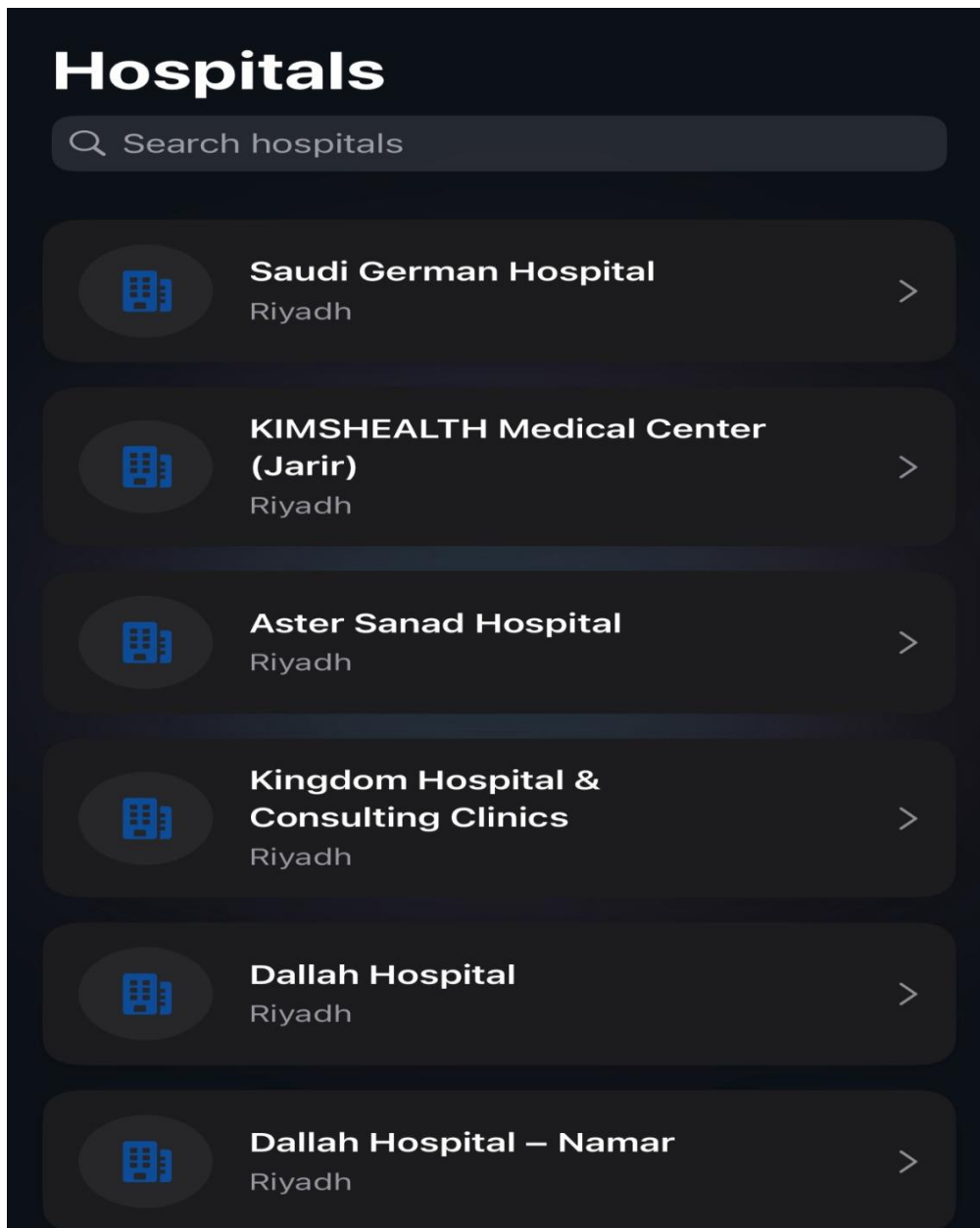
Testing was conducted throughout the development process to ensure system reliability and correctness. Unit testing was used to verify individual system components, while integration testing ensured proper interaction between system modules. Functional testing validated that core features such as appointment booking and data access operated as intended. User-based evaluation was also considered to assess system usability and overall patient experience.

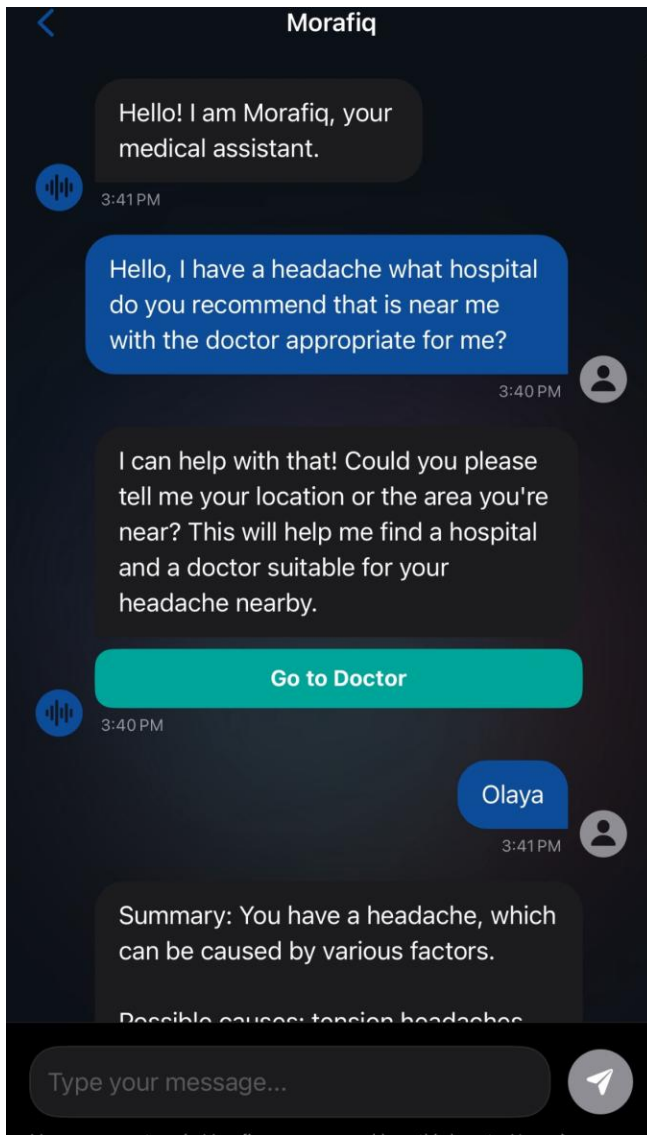
3.6 Ethical, Legal & Security Considerations

The project addresses ethical and legal considerations related to patient data privacy and confidentiality. The system design emphasizes secure authentication, controlled access to information, and protection of sensitive healthcare data. Ethical guidelines were followed to ensure responsible use of AI-based(Morafiq) recommendations, with the system designed to support and not replace professional medical judgment. Security considerations include data encryption, secure communication, and compliance with relevant healthcare data protection regulations.

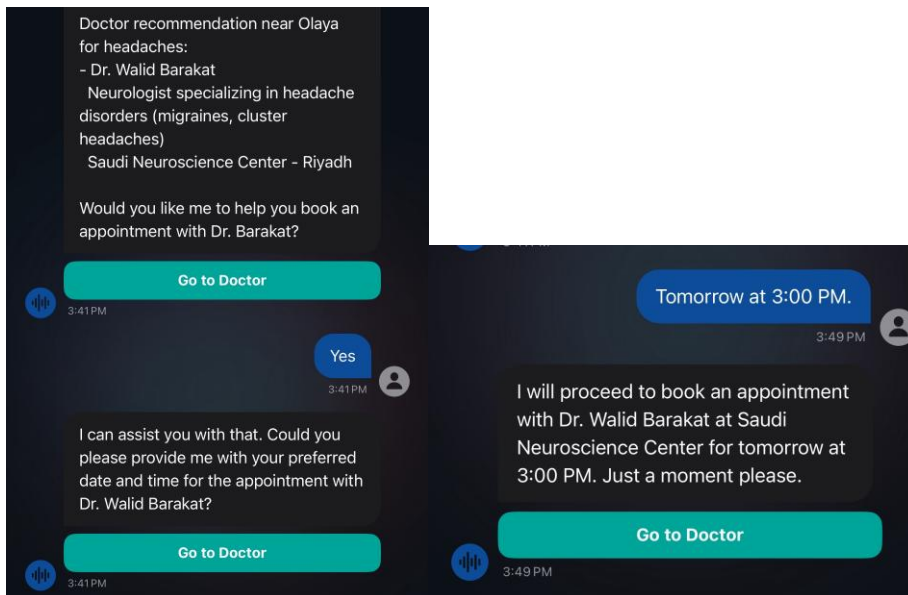
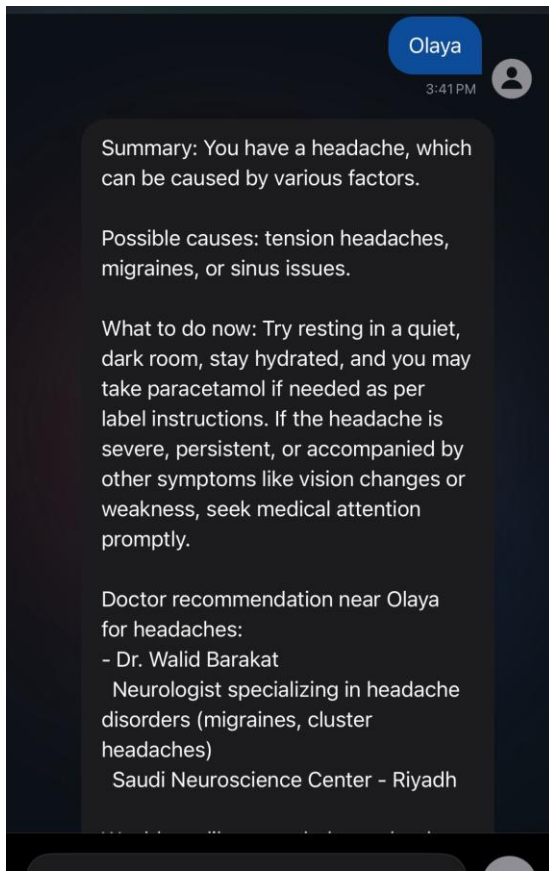


Hayat Mobile Application Implementation With Morafiq:





Using Morafiq to Book an Appointment



Successfully Booked.

4.2 Evaluation

The evaluation of the system focused on three key areas: functionality, usability, and security. Testing confirmed the application's successful support for core features, specifically appointment booking and the delivery of AI-based recommendations. Furthermore, user feedback highlights a positive impact, noting improvements in both efficiency and ease of use.

4.3 Individual and Group Contributions

The team members each had specialized roles: Ahmed Almuhanha focused on mobile UI/UX design; Adam Aljuwayed managed security and privacy engineering; Abdulaziz Alabdullatif developed the AI logic and integration; and Nawaf Aljarrah was responsible for project documentation. In addition to these specific duties, every member collaborated on requirements analysis, system design, testing, and the final presentation.

Project Tasks & Task Assignments

TASK	DETAILS	ASSIGNED TEAM MEMBERS
PROJECT SETUP	Initial project planning, team coordination, defining scope, identifying objectives, setting milestones, and preparing a	ALL TEAM MEMBERS

	development environment.	
PROJECT PROPOSAL	defining problem statements, objectives, expected deliverables, risk assessment, and software process selection.	All Team Members
REQUIREMENTS GATHERING AND ANALYSIS	Identifying functional and non-functional requirements, analyzing existing healthcare applications, defining system scope, and documenting requirements specifications.	Nawaf
SYSTEM DESIGN	Designing system architecture, defining system components, preparing UML diagrams and data flow structure, and defining AI integration approach.	Abdulaziz
DEVELOPMENT	Frontend mobile application	Adam, Ahmed

	development, backend logic implementation, integration of AI-based symptom analysis, and database configuration.	
TESTING & VALIDATION	Conducting unit testing, integration testing, functional validation, security checks, and verifying system usability and performance.	Ahmed, Adam, Nawaf
DEPLOYMENT	Preparing system for demonstration, configuring environment for prototype presentation, and ensuring system stability during final demo	Ahmed
FINAL REPORT & PRESENTATION	Preparation of final IS499 report, formatting according to template, compiling documentation, and preparing presentation slides for project defense.	Nawaf (Form Documentation) ALL team members (Presentation)

Chapter 5 – Conclusion and Future Work

Hayat is a unified hospital application developed to tackle the fragmented nature of the Saudi Arabian healthcare system. The project successfully demonstrated a single platform integrating various healthcare services and incorporating AI such as Morafiq for doctor recommendations, thus improving healthcare accessibility and the patient experience through the strategic use of Information Systems. Future endeavors will focus on integrating with real hospital APIs, refining the AI's accuracy with authentic medical data, and scaling the deployment nationwide.

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